LAB 07 FUZZ TESTING SE LAB

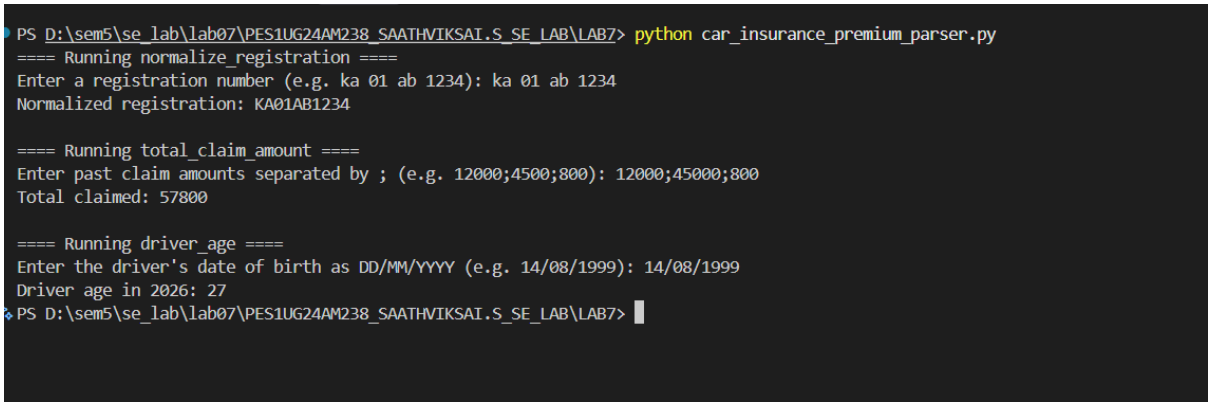
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AIML SEC-D

SCREENSHOTS

1)Run the Buggy Application



2) Failing test cases/errors produced by Hypothesis

```
PS D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7> python -m pytest test_car_insurance_premium_parser.py
============================= test session starts ==============================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7
plugins: hypothesis-6.170.15, platformdirs-4.13.6
collected 3 items

test_car_insurance_premium_parser.py FFF                                 [100%]

=================================== FAILURES ===================================
_____________________ test_normalize_registration_no_crash _____________________

s = None

    @given(st.text() | st.none())
    def test_normalize_registration_no_crash(s):
        try:
>           normalize_registration(s)

test_car_insurance_premium_parser.py:7:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

plate = None

    def normalize_registration(plate):
        """
        Removes spaces and dashes from a registration number and upper-cases it.
        Assumes plate is a non-empty string.
        """
>       plate = plate.strip().upper()
                ^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'

car_insurance_premium_parser.py:6: AttributeError
```

```
During handling of the above exception, another exception occurred:

    @given(st.text() | st.none())
>   def test_normalize_registration_no_crash(s):
                ^^^

test_car_insurance_premium_parser.py:5:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = None

    @given(st.text() | st.none())
    def test_normalize_registration_no_crash(s):
        try:
            normalize_registration(s)
        except Exception as e:
            # If the function crashes on unhandled inputs, fail the test and show what caused it
>           assert False, f"normalize_registration crashed with: {e}"
E           AssertionError: normalize_registration crashed with: 'NoneType' object has no attribute 'strip'
E           assert False
E           Failing test case: test_normalize_registration_no_crash(
E               s=None,
E           )

test_car_insurance_premium_parser.py:10: AssertionError
_____________________ test_total_claim_amount_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 13, in test_total_claim_amount_no_crash
  |     def test_total_claim_amount_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 15, in test_total_claim_amount_no_crash
```

```
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 22, in test_driver_age_no_crash
    |     driver_age(s)
    |     ^^^^^^^^^^^^^
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\car_insurance_premium_parser.py", line 23, in driver_age
    |     return current_year - int(dob.split('/')[2])
    |                           ^^^^^^^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 24, in test_driver_age_no_crash
    |     assert False, f"driver_age crashed with: {e}"
    | AssertionError: driver_age crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_driver_age_no_crash(
    |     s=None,
    | )
    +------------------------------------
=========================== short test summary info ============================
FAILED test_car_insurance_premium_parser.py::test_normalize_registration_no_crash - AssertionError: normalize_registration crashed with: 'NoneType' object has no attribute 'strip'
FAILED test_car_insurance_premium_parser.py::test_total_claim_amount_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_car_insurance_premium_parser.py::test_driver_age_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
============================== 3 failed in 2.70s ===============================
PS D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7>
```

```
    |     total_claim_amount(s)
    |     ~~~~~~~~~~~~~~~~~~~^^^
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\car_insurance_premium_parser.py", line 16, in total_claim_amount
    |     return sum(int(amount) for amount in text.split(';'))
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\car_insurance_premium_parser.py", line 16, in <genexpr>
    |     return sum(int(amount) for amount in text.split(';'))
    |                ~~~^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 17, in test_total_claim_amount_no_crash
    |     assert False, f"total_claim_amount crashed with: {e}"
    | AssertionError: total_claim_amount crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_total_claim_amount_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 15, in test_total_claim_amount_no_crash
    |     total_claim_amount(s)
    |     ~~~~~~~~~~~~~~~~~~~^^^
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\car_insurance_premium_parser.py", line 16, in total_claim_amount
    |     return sum(int(amount) for amount in text.split(';'))
    |                                           ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 17, in test_total_claim_amount_no_crash
    |     assert False, f"total_claim_amount crashed with: {e}"
```

```
    | AssertionError: total_claim_amount crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_total_claim_amount_no_crash(
    |     s=None,
    | )
    +------------------------------------
_____ test_driver_age_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 20, in test_driver_age_no_crash
  |     def test_driver_age_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 22, in test_driver_age_no_crash
    |     driver_age(s)
    |     ~~~~~~~~~~^^^
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\car_insurance_premium_parser.py", line 23, in driver_age
    |     return current_year - int(dob.split('/')[2])
    |                           ~~~~~~~~~~~~~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7\test_car_insurance_premium_parser.py", line 24, in test_driver_age_no_crash
    |     assert False, f"driver_age crashed with: {e}"
    | AssertionError: driver_age crashed with: list index out of range
    | assert False
    | Failing test case: test_driver_age_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
```

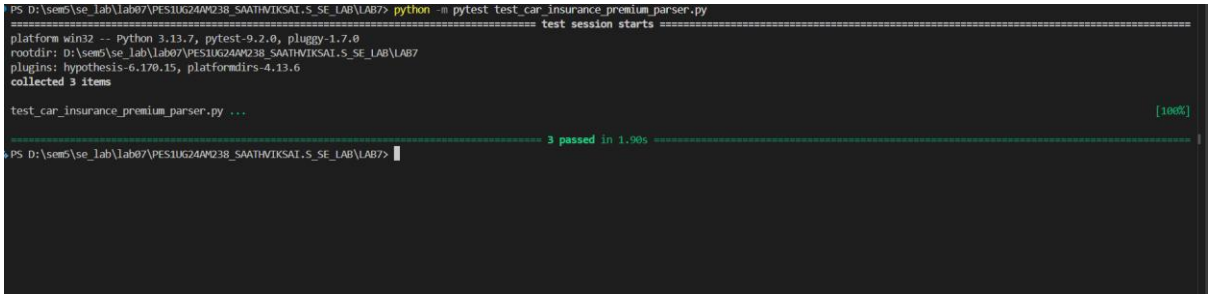
3)After fixing the bugs in car_insurance_premium.py

```
PS D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7> python car_insurance_premium_parser.py
==== Running normalize_registration ====
Enter a registration number (e.g. ka 01 ab 1234): ka 01 ab 2345
Normalized registration: KA 01 AB 2345

==== Running total_claim_amount ====
Enter past claim amounts separated by ; (e.g. 12000;4500;800): 12000;67000;800
Total claimed: 79800

==== Running driver_age ====
Enter the driver's date of birth as DD/MM/YYYY (e.g. 14/08/1999): 14/02/2006
Driver age in 2026: 20
PS D:\sem5\se_lab\lab07\PES1UG24AM238_SAATHVIKSAI.S_SE_LAB\LAB7>
```

4) After fixing the test file



- What is Fuzzing?
  Fuzzing (or fuzz testing) is a dynamic testing technique that feeds random, unexpected, or malformed data into a program to uncover bugs, crashes, and security vulnerabilities, and to test its robustness against edge cases.
- Why Use Fuzzing?
  Manual and traditional unit tests often miss rare edge cases or focus only on anticipated failures. Fuzzers automate test generation to uncover problems you may not have considered, including unexpected exceptions, incorrect behaviour, and security flaws.
- What is Hypothesis?
  Hypothesis is a property-based fuzz-testing library for Python. Rather than manually writing individual test cases, you describe the general properties your function should satisfy (for example, that it should not crash). Hypothesis then automatically generates dozens or hundreds of random inputs using strategies to check those properties and minimise any failing inputs. s.